Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (EC) Examination

March-April 2018

EC302.01/EC302 Integrated Circuits and Applications

Date: 29.03.2018, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the question below.** **[07]**

a. State various applications of Sample and Hold circuit. [02]

b. Draw a circuit diagram for negative clipper using IC741. [02]

c. Draw a circuit diagram for non inverting amplifier using IC741 also write gain equation. [02]

d. The change in output voltage for the corresponding change in load current in a 7805 IC regulator is defined as____________. [01]

   A. All of the mentioned
   B. Load regulation
   C. Line regulation
   D. Input regulation

Q – 2.a List the benefits of switching regulators. Draw block diagram of switching regulator and explain it in detail. **[04]**

Q – 2.b **Answer any two.** **[10]**

a. Design absolute wave output circuit with necessary diagram and mathematical expression.

b. Explain Basic Non inverting comparator with appropriate circuit diagram and waveforms.

c. Explain Zero crossing detector with appropriate circuit diagram and waveforms.

Q - 3 **Answer any two.** **[14]**

a. Design an astable multivibrator using IC555 which will generate square wave with a frequency of 200Hz with a duty cycle of 78%.

b. Design PLL using IC 565. Find the free running frequency, Lock range and capture range if R1= 10KΩ, C1=0.01μF, C2=10μF and V=+10 Volt.

c. Design a Practical Integrator with a DC gain of 10, to integrate a square wave of 10KHz.

SECTION – II

Q - 4 Answer the question below. **[07]**

a. Which of the following transfer functions cannot be realized in the real world and are unstable ? [01]

P: $\frac{s+2}{s^2+5s+6}$ Q: $\frac{s-2}{s^2+5s+6}$ R : $\frac{s+2}{s^2-5s+6}$ S: $\frac{s-2}{s^2-5s+6}$

A. Only P
B. Only Q
C. Only Q& R
D. Only R & S

b. Select the correct relation for the mentioned attenuation parameters of a Band stop filter? [01]

α_{max} : Maximum Attenuation in Passband
α_{min} : Minimum Attenuation in Stopband

A. $\alpha_{max} < \alpha_{min}$
B. $\alpha_{max} > \alpha_{min}$
C. $\alpha_{max} = \alpha_{min}$
D. None of these

c. In filter design quality factor is desired to be greater than 1. Which of the following is true for location of poles in complex plane? (Hint: $\Psi = \cos^{-1}(1/2Q)$)? [01]

A. $\Psi < 45°$ B. $\Psi = 90°$
C. $\Psi < 60°$ D. $\Psi > 60°$

d. What is the sequence of tuning the parameters of filters while employing orthogonal tuning? [01]

A. $w_0 \rightarrow H \rightarrow Q$ B. $w_0 \rightarrow Q \rightarrow H$
C. $Q \rightarrow H \rightarrow w_0$ D. None of these

e. Define following terms: [03]

1) Input offset current
2) CMRR
3) Power supply rejection ratio (PSRR)

Q – 5.a What is op-amp? Discuss ideal OP-AMP electrical characteristics and ideal voltage transfer curve. **[05]**

OR

Q – 5.a The 741 op-amp having the following parameters is connected as a Voltage follower**:** A = 2, 00,000,Ri= 2 MΩ, Ro= 75Ω, fo= 5 Hz, $+V_{CC}$= +15 V,$-V_{EE}$= -15 V, and Output voltage swing = ± 13 V. Compute the values of A_F, R_{iF}, R_{oF} , f_F and V_{ooT}. **[05]**

Q – 5.b Determine the output voltage in open loop differential amplifier if, **[04]**
a. Vin1 = 5 μ V dc , Vin2 = -7 μ V dc ,
b. Vin1 = 10 mV rms , Vin2 = 20 mV rms ,
The op-amp is 741 with the following specifications: A = 2,00,000 , Ri= 2 MΩ , Ro= 75Ω, +VCC= +15 V ,-VEE= -15
V, and Output voltage swing = ± 14 V.

Q – 5.c Design and Discuss application of OP-AMP as open loop configuration. Also define limitations of open loop configuration. **[05]**

OR

Q – 5.c Explain types of differential amplifier and important characteristics of differential amplifier. **[05]**

Q – 6 **Answer any two.** **[14]**

a. Design a first order high pass filter such that its dc gain T(0) = 0.3 and the gain at high frequencies T(∞) = 1. The filter must have a zero on the negative real axis at f_Z = -159 Hz. Do the following for this filter : [07]

A. Draw the circuit for the High pass filter mentioned.
B. Values of the component of the circuit (obtained in (a))
C. Pole Zero location of this filter
D. Transfer function of the high pass filter

b. Draw Sallen-Key circuit low pass filter. Derive it's transfer function. [07]

c. Realize a circuit which gives butterworth response having following values: α_{max}=0.5 dB , α_{min} =20 dB, w_s=2000 rad/sec , w_p=1000 rad/sec. [07]

d. Explain the following terms. α_{max}, α_{min}, Transition band and hence draw the attenuation characteristics of Low Pass Filter, High Pass Filter, Band Pass filter, Band Stop Filter. [07]

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